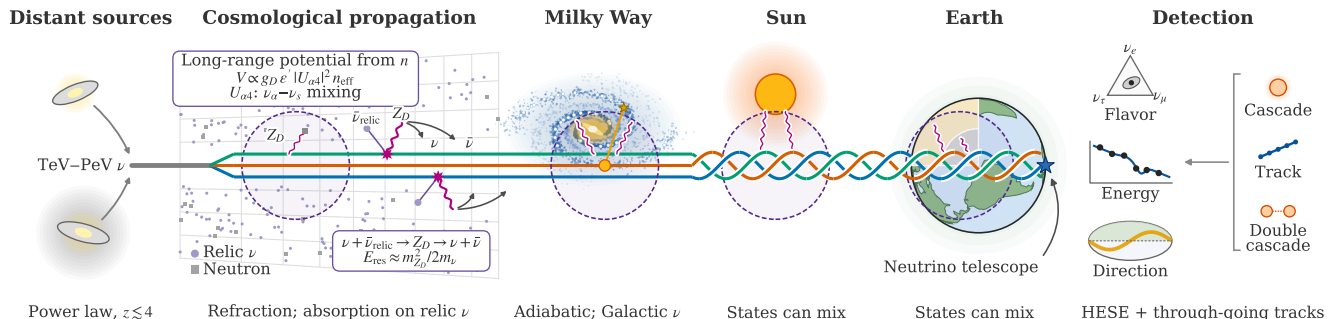

Distant sources
Cosmological propagation
Milky Way
Sun
Earth
Detection
Long-range potential from n
$V \propto g_D\, e^{-} |U_{a4}|^2\, n_{\rm eff}$
U_{a4}: ν_a–ν_s mixing
TeV–PeV ν
Z_D
$\bar{\nu}_{\rm relic}$
ν
$\bar{\nu}$
Relic ν
Neutron
$\nu + \bar{\nu}_{\rm relic} \to Z_D \to \nu + \bar{\nu}$
$E_{\rm res} \approx m_{Z_D}^2 / 2 m_\nu$
Neutrino telescope
ν_e
ν_τ
ν_μ
Flavor
Energy
Direction
Cascade
Track
Double cascade
Power law, $z \lesssim 4$
Refraction; absorption on relic ν
Adiabatic; Galactic ν
States can mix
States can mix
HESE + through-going tracks